



Lab Manual

DatabaseSystemLab

(CC-215-L)

Faculty of Computing & Information Technology
(FCIT) University of the Punjab, Lahore.

www.pucit.edu.pk

CC-215-L Database Systems Lab

Lab#	Topics
1	Installation of Oracle / MS SQL Server, Introduction to SQL Commands

2	Data Retrieval, SELECT statement, Arithmetic Operators, Relational Operators, Logical Operators, WHERE clause
3	Special operators (BETWEEN, IN, LIKE, IS NULL), Use of DISTINCT, Column Alias, ORDER BY clause
4	SQL Functions, Single row Functions, Handling NULL values (NVL), Character Functions
5	Functions for numeric and date data types, Type conversion Functions
6	Decode and CASE expressions, Mult-row (group) Functions, Group by clause, Having clause
7	Revision of the above topics, Use of ROLLUP, CUBE, Introduction to SQL Joins
8	SQL Join Types: Cartesian Join, Inner or Equi Join, Outer Join (Left, Right, Full), Non-Equi Join, Self-Join
9	Subqueries, Single-row and Multiple-row subqueries, Use of IN, ANY, ALL operators in subqueries, Multiple-Column Subqueries, Correlated Subquery,
10	DDL commands, Creating database tables, Applying table and column level constraints (PRIMARY KEY, NOT NULL, UNIQUE, FOREIGN KEY, CHECK, DEFAULT), ALTER command, Data Dictionary views for tables and constraints management, Disabling and Enabling of constraints, Renaming a table or column, Drop, rename, and truncate tables,
11	Creating a table using subqueries, DML commands (INSERT, UPDATE, DELETE), Virtual Tables (Views), Indexes, Sequence, Synonyms
12	Revision, Controlling User Rights & Access (DCL commands), Role, Privileges, Creating a database user, GRANT, REVOKE
13	Introduction to Procedural SQL (PL/SQL), Sections of a PL/SQL block, Variable declaration and initialization, SELECT statement in PL/SQL, Arithmetic expressions
14	PL/SQL: Selection, Repetition, Exception Handling, Cursors
15	PL/SQL: Stored Procedures and Functions, Introduction to Database Triggers
16	Database Triggers, Revision

CC-215-L

Database Systems Lab

Laboratory 01

Installation of Oracle / MS SQL Server &

Introduction to SQL & Basic SQL Queries

Version: 1.0.1

Release Date: 10-09-2026

**Department of Information
Technology University of the Punjab
Lahore, Pakistan**

Contents

Topics	0
Learning Objectives	0
General Instructions:	0
Installation of Oracle / MS SQL Server	1
SQL Background and Overview	11
Introduction to SQL Statements	11
Writing SQL Statements	11
SELECT Statement	12
Examples of basic SELECT statements	12
Concatenation Operator ()	12
Eliminating Duplicate Rows	13
References and Additional Material	13

Topics

- Installation of Oracle / MS SQL Server
- Use of Oracle 11g / MS SQL Server
- Setup Apex Oracle as an alternative source
- Introduction to SQL Commands
- SELECT statement

Learning Objectives:

- Understand the basic steps of Oracle / MS SQL Server installation
- Understand the Oracle / MS SQL Server environment
- Understanding about SQL
- Writing Basic Queries for data retrieval

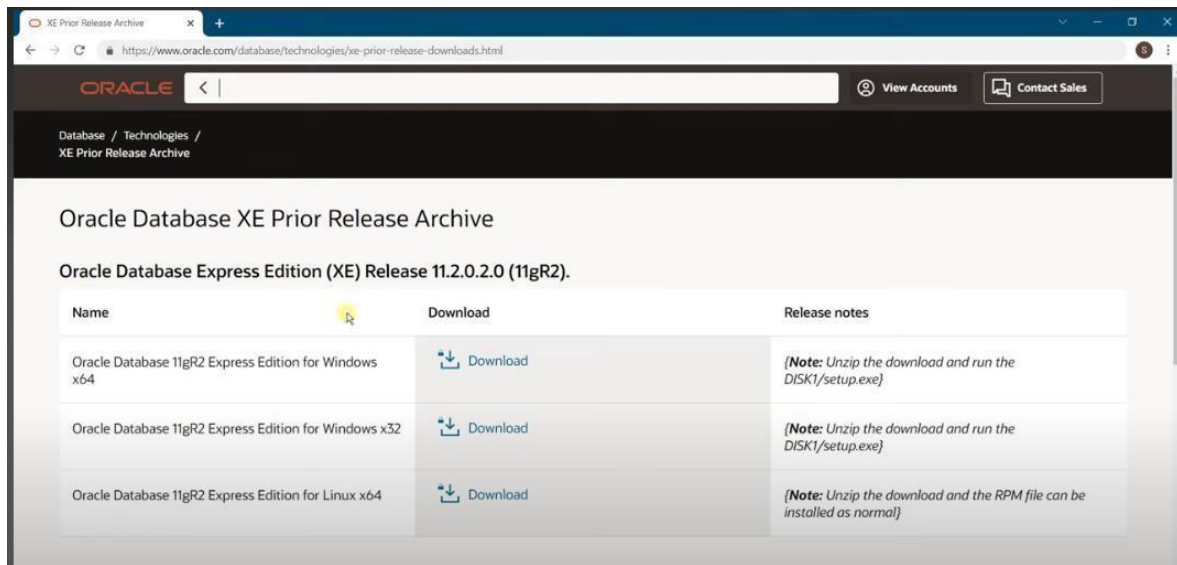
Lab Instructors:

Course Instructor	Ahmad Hassan ahmad.hassan@pucit.edu.pk
Teaching Assistants	Muhamad Mudassir Baig
	Maryam Raza
	Supporting TAs: Maaz Bin Asif , Zohaib
	Saad , Abdul Rehman , Sultan

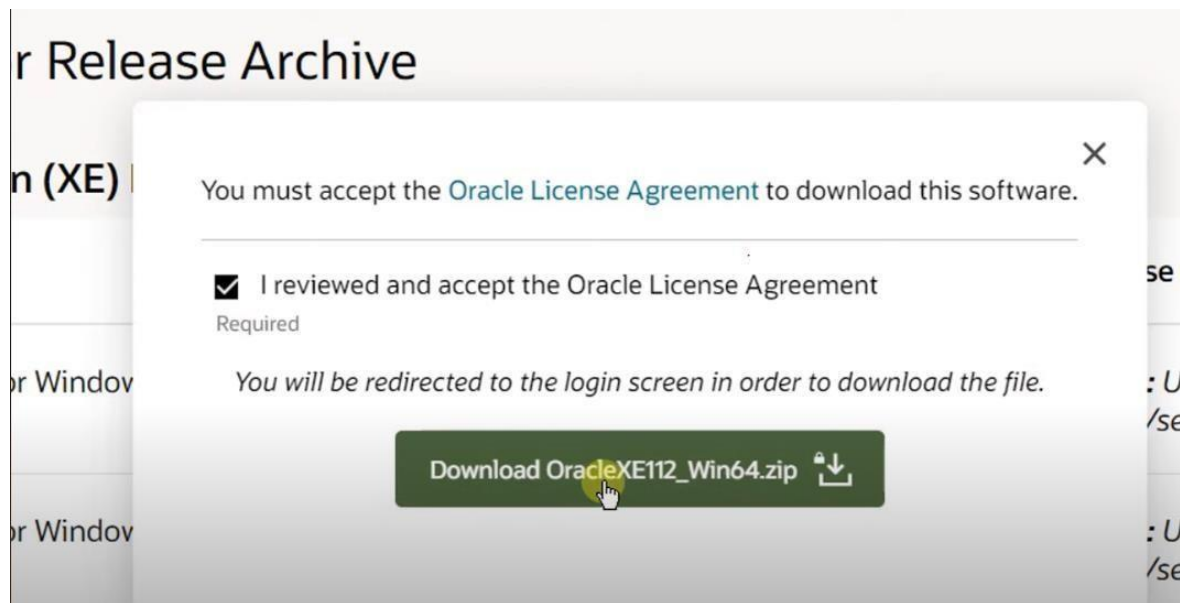
Lab-01

Installation of Oracle / MS SQL Server:**Step 1:**

Go to Oracle official site (<https://www.oracle.com/database/technologies/xe-prior-releasedownloads.html>) and download the Oracle Database Express Edition (XE) edition for Windows

**Step 2:**

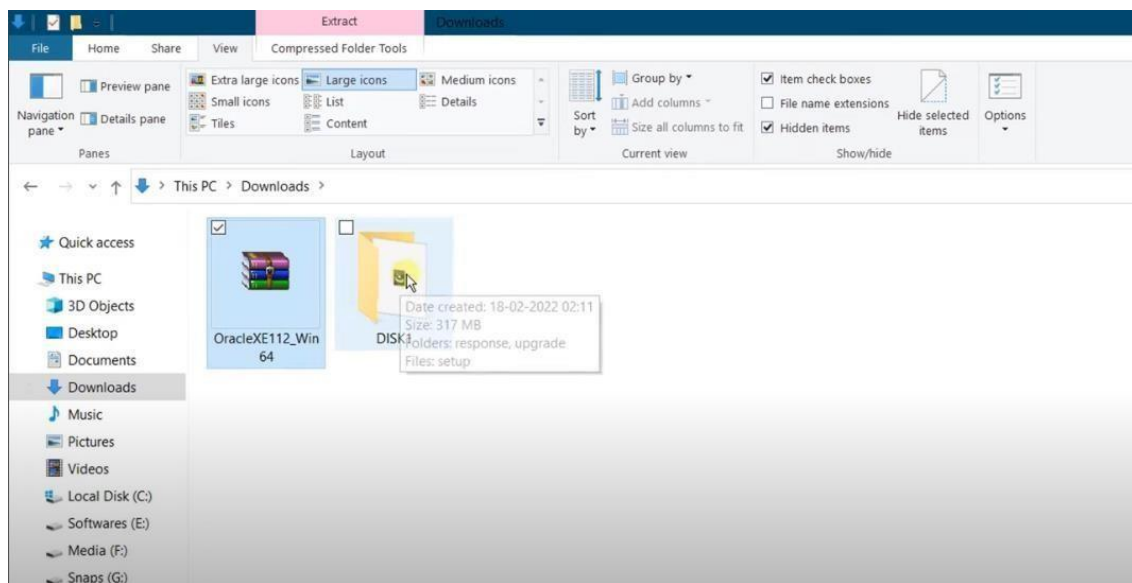
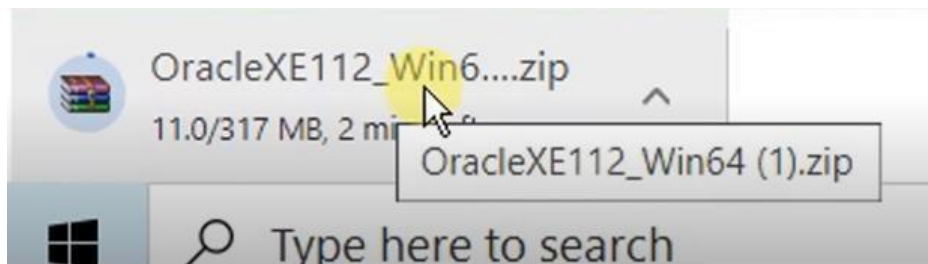
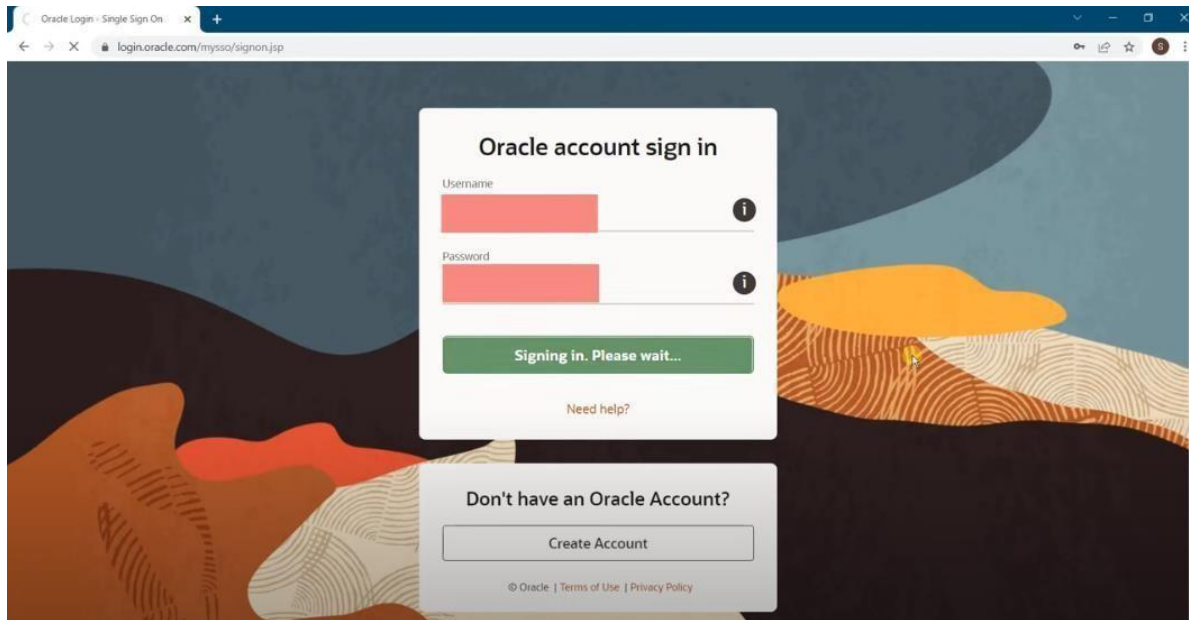
Click on the Download button based on the version you want to download accept the Oracle License agreement.



Lab-01

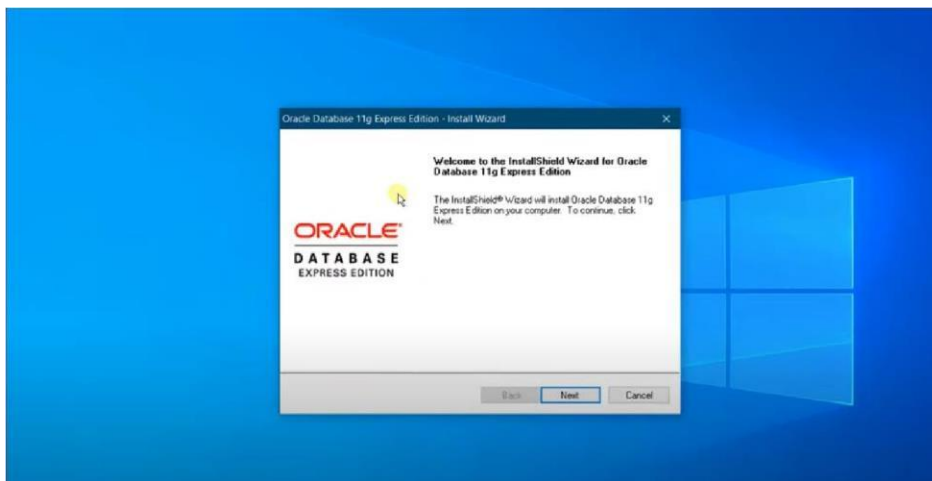
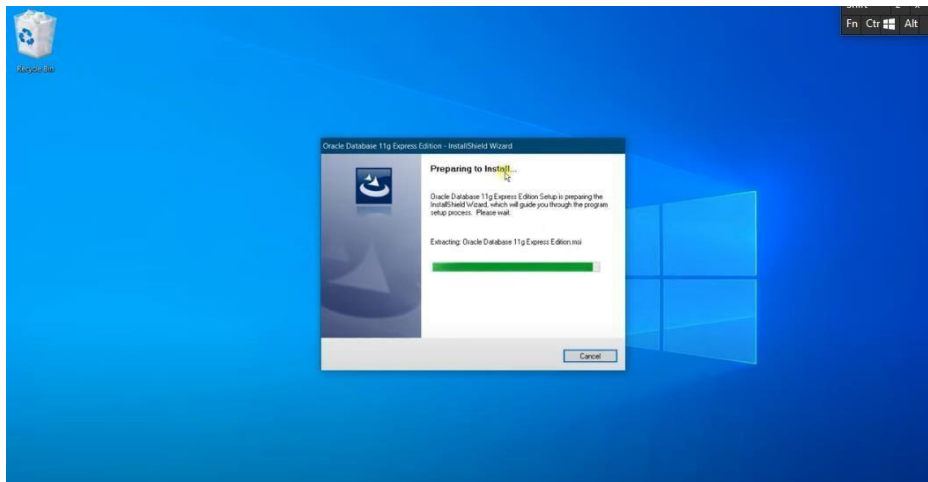
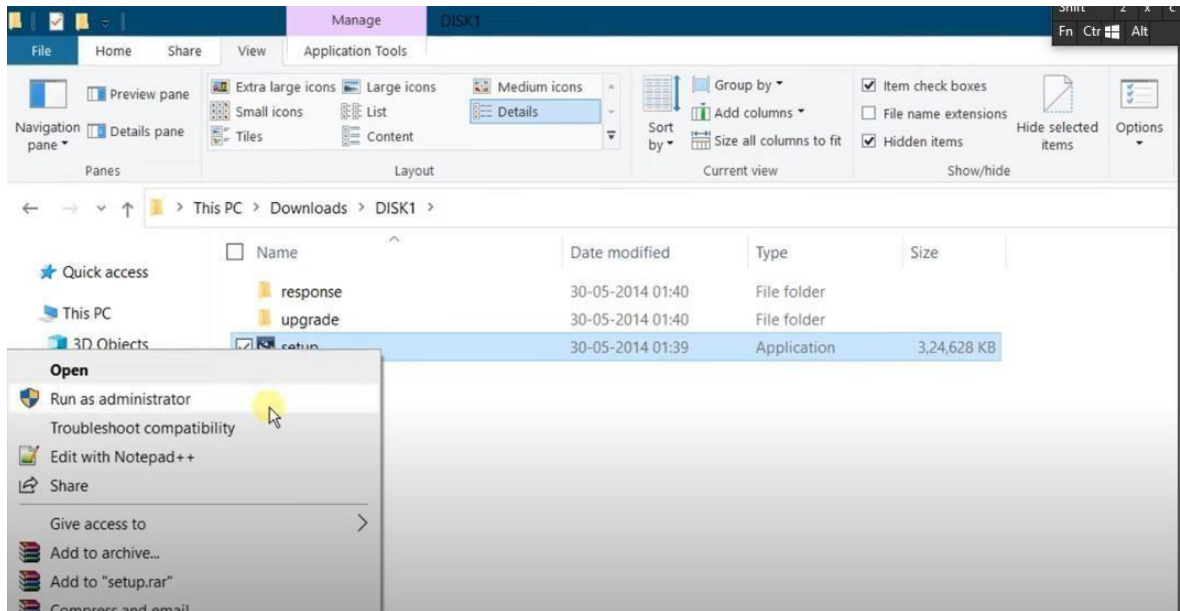
Step 3:

Enter your Oracle account to start the download. If you don't have any, please create one by using a Gmail account.



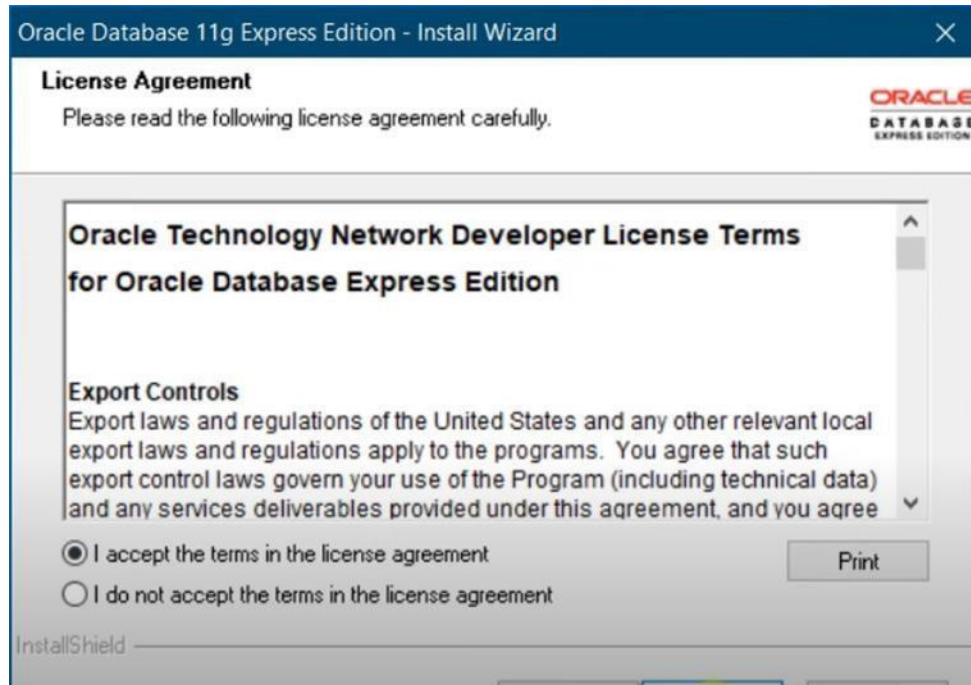
Step 4:

Extract these files and you can see the DISK named Folder there please run the setup.exe file present in this folder.



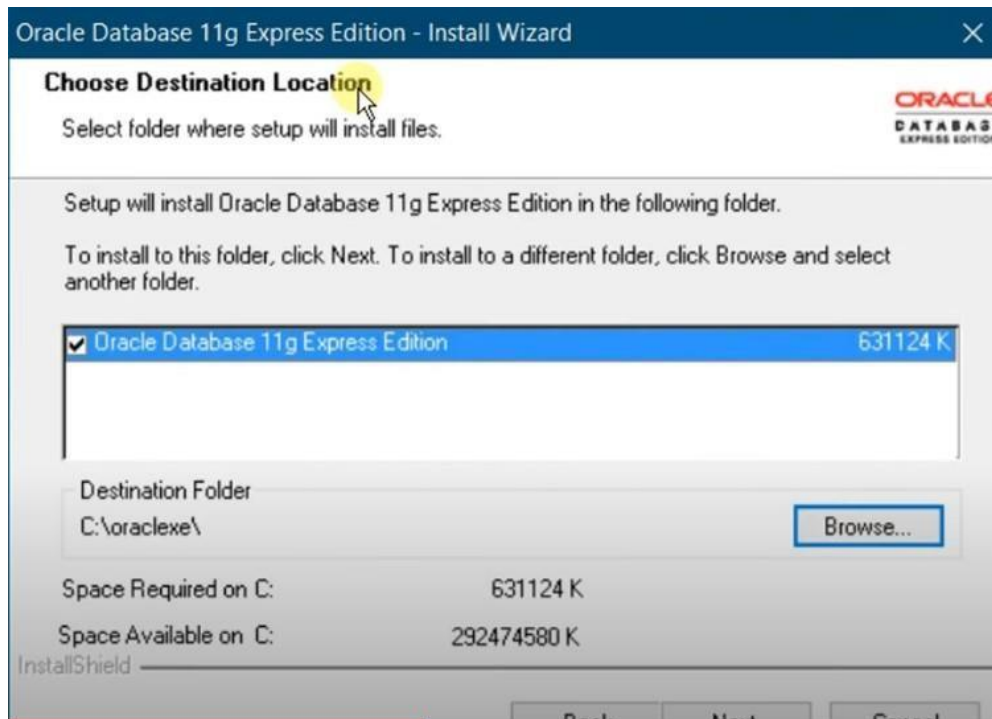
Step 5:

Accept the license agreement from here as well and click next



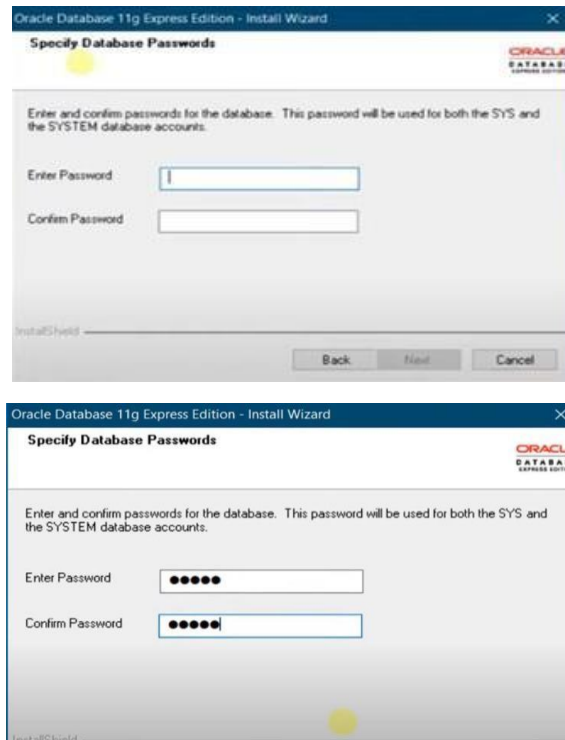
Step 6:

Select the path of folder where you want to install the oracle 11g setup.



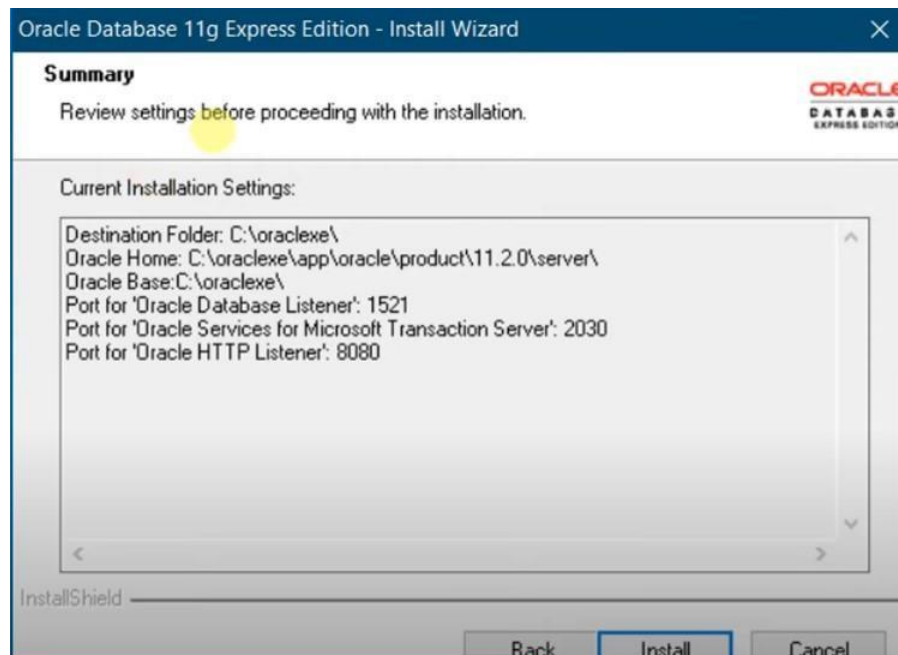
Step 7:

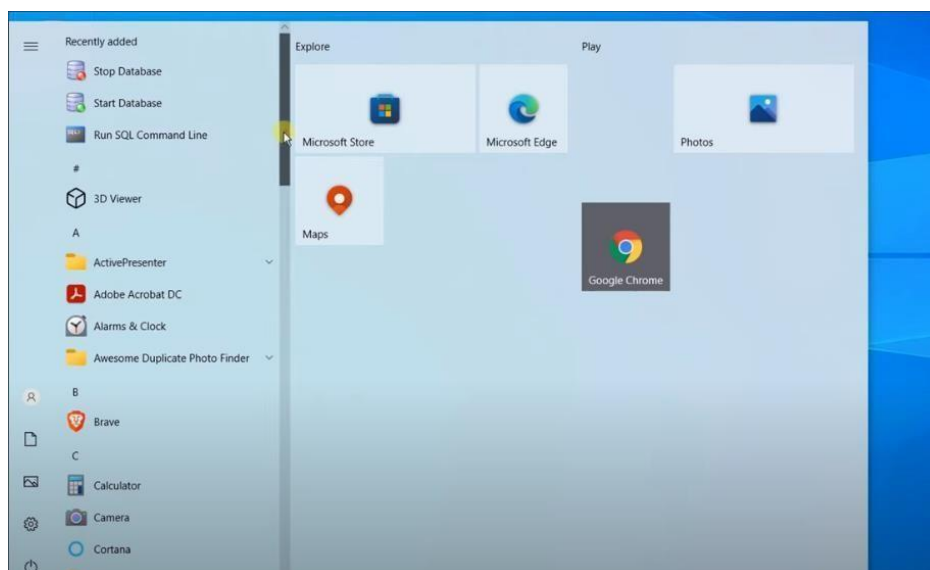
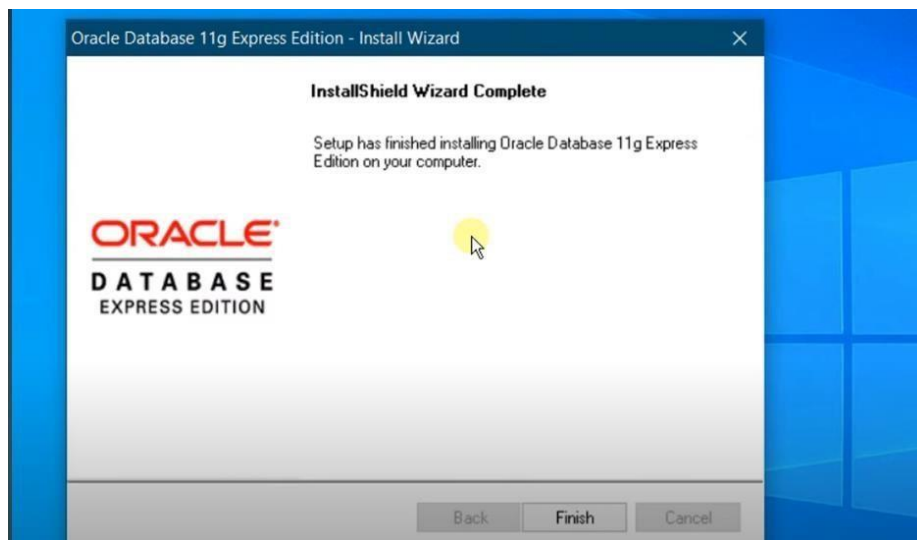
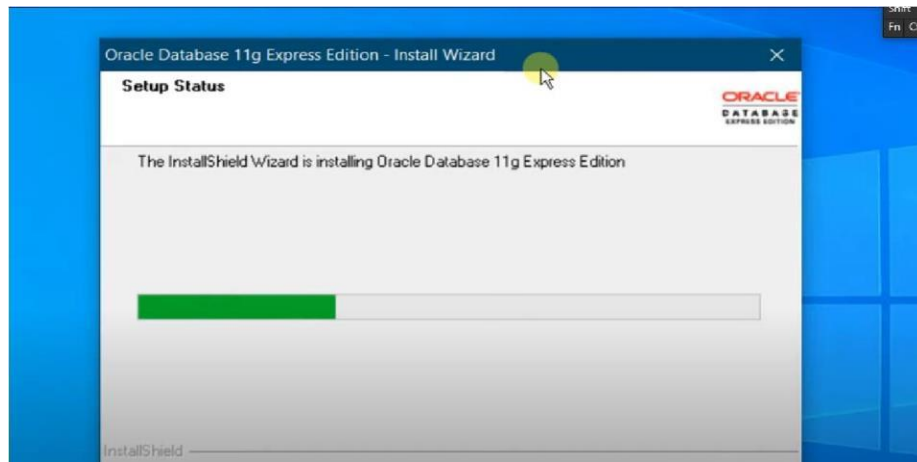
Enter the by default password “admin” here.

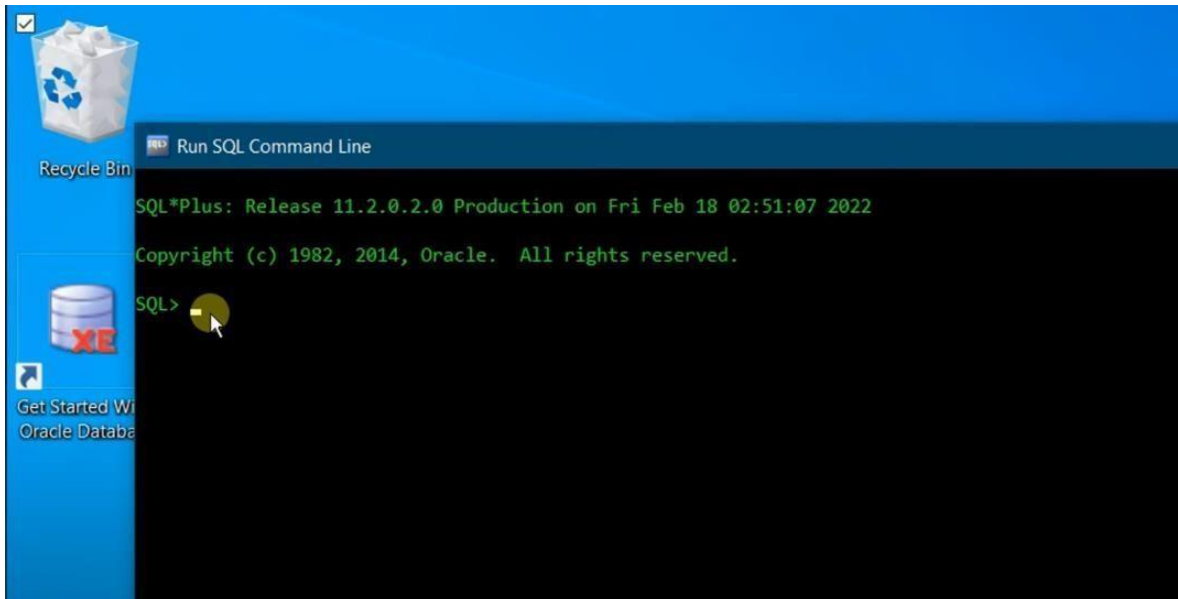


Step 8:

Have quick look to summary and click next to where your installation will completed.

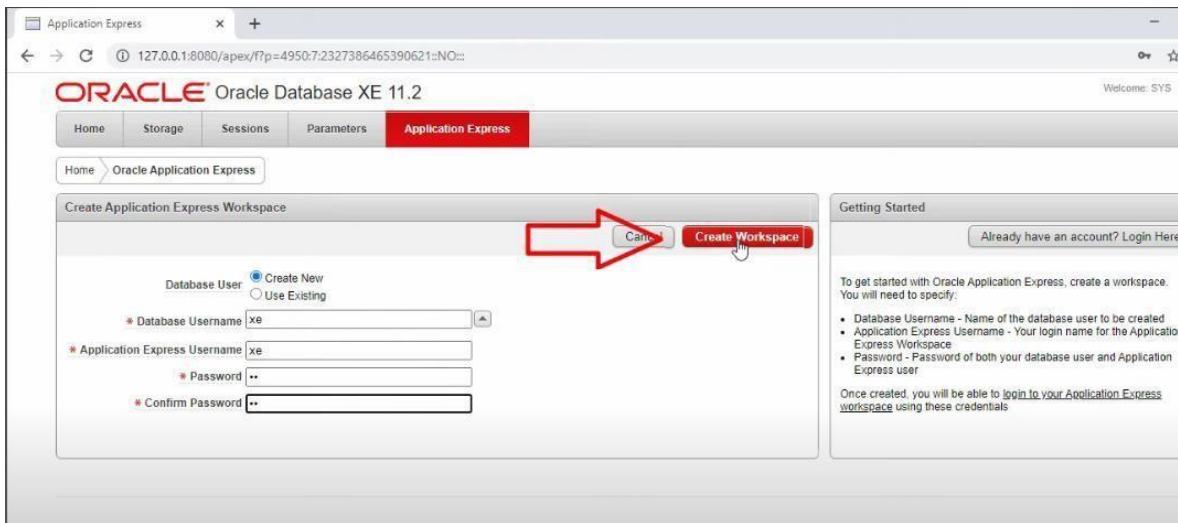






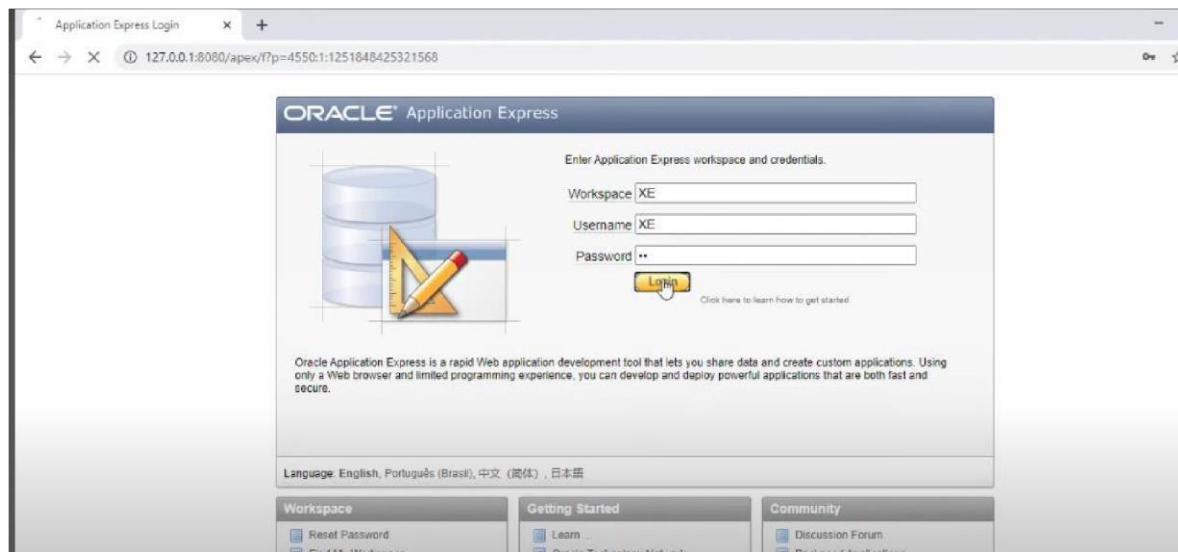
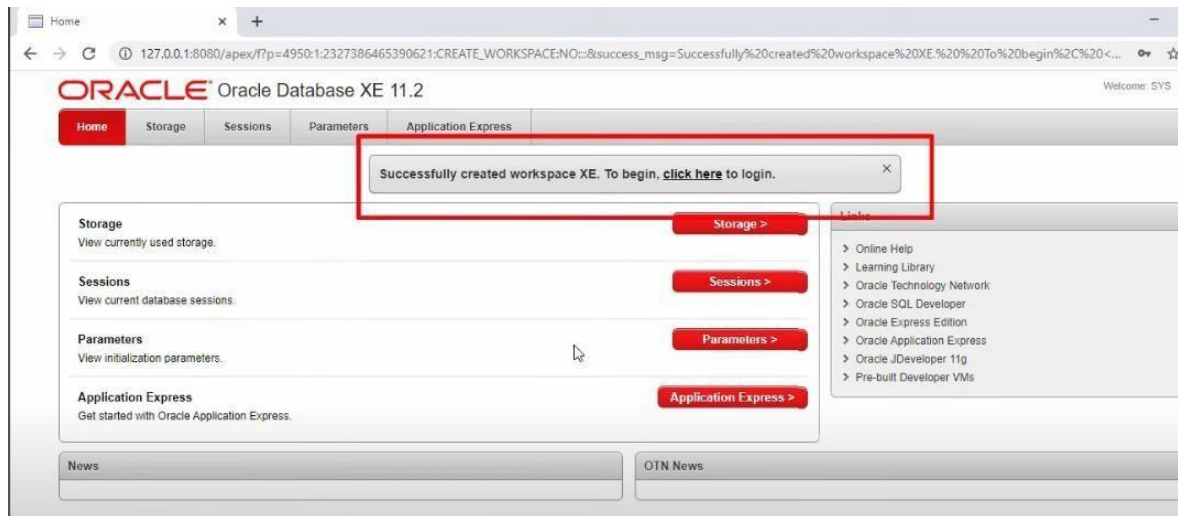
Step 9:

Open the icon for oracle 11g from desktop and First enter the password username as “sys” and password “admin” then you will get this screen. Please Create a new workspace here for you.



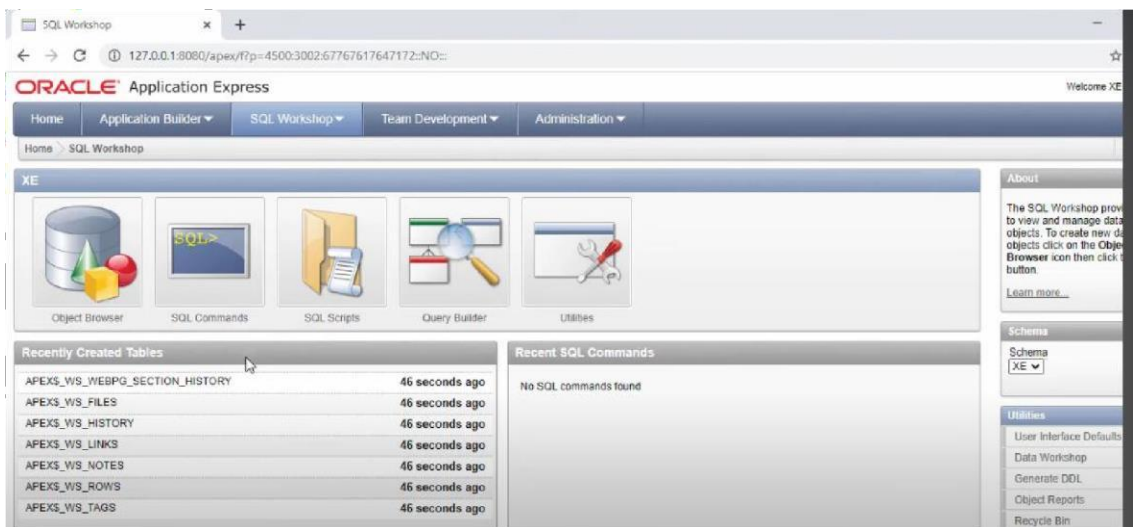
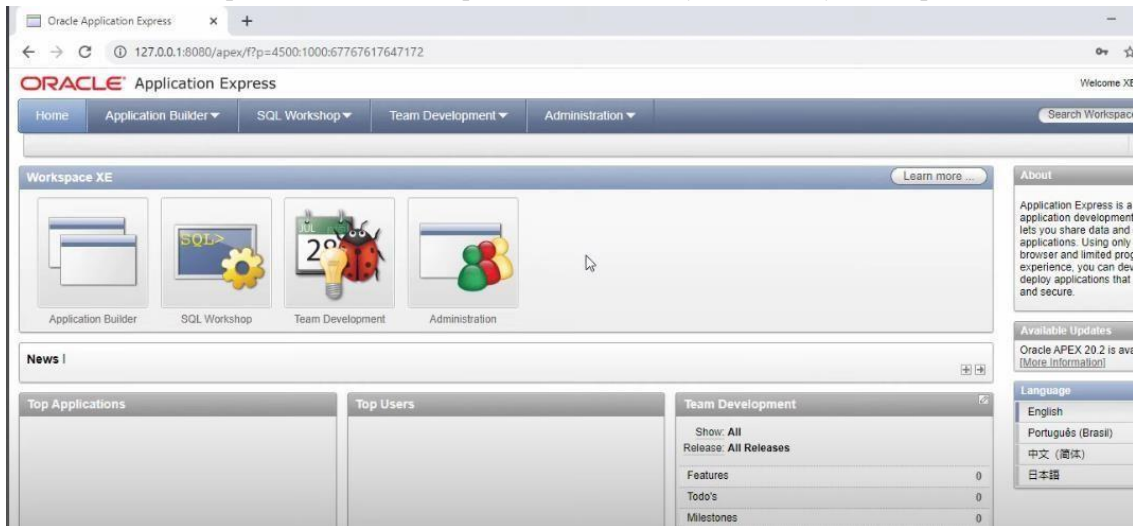
Step 10:

After the Workshop has been created successfully, login to this by putting username and password.

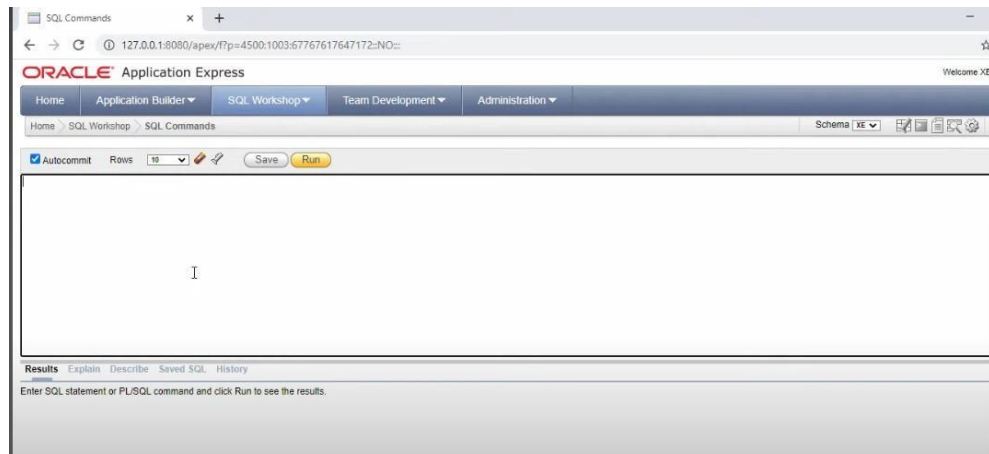


Step 11:

Go to SQL workshop and then towards sql commands where you can run your scripts.

**Step 12:**

Now you can easily run your sql queries for Lab here in this section.



SQL Background and Overview:

- SQL pronounced as Sequel is a language for RDBMS or SQL makes RDBMS possible.
- It was first defined by Chamberlin and others at IBM's research laboratory in San Jose, California (Late 1970's)
- ANSI and ISO have worked for the standardization of SQL.
- SQL is a non-procedural language.
- SQL does much more and is more powerful than just asking queries from the database.
 - Using SQL, one can:
 - Create Relations or Tables.
 - Insert, Delete and Update Records.
 - Apply Validation Checks.
 - Create users.
 - Apply Security Restrictions etc.

Introduction to SQL Statements

The categories of SQL statements are given in the following table

Category / Language	Statements / Commands
1. Data Retrieval	• Select
2. Data Manipulation Language (DML)	• Insert • Delete • Update
3. Data Definition Language (DDL)	• Create • Alter • Drop • Rename • Truncate
4. Data Control Language (DCL)	• Grant • Revoke
5. Transaction Control Language (TCL)	• Commit • Rollback • Save point

Writing SQL Statements

- SQL statements are not case sensitive.
- SQL statements can be on one or more lines.
- Keywords cannot be abbreviated or split across lines.
- Clauses are usually placed on separate lines.
- Tabs and indents are used to enhance readability.

SELECT Statement

- The Select statement is the building block for querying the tables:
- Syntax:

```
SELECT [DISTINCT] { * | column [alias],...}  
FROM table (s)  
[WHERE clause]  
[ORDER BY clause]  
[GROUP BY clause]  
[HAVING clause]
```
- SELECT identifies *what* columns
- FROM identifies *which* table
- WHERE clause is used for filtering the rows of the SELECT statement
- ORDER BY clause is used for sorting the output on the basis of certain column(s)
- HAVING clause is used writing a condition that uses group or aggregate functions
- The following tables will be used for practicing the SELECT statement

EMP

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7839	KING	PRESIDENT		17-NOV-81	5000		10
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
			7698	08-SEP-81	1500	0	30
			7698	03-DEC-81			
			7698	22-FEB-81			
			7566	03-DEC-81			
			7902	17-DEC-80			
			7566	09-DEC-82			
			7788	12-JAN-83			
7934	MILLER	CLERK	7782	23-JAN-82			

DEPT

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

SALGRADE

GRADE	LOSAL	HISAL
1	700	1200
2	1201	1400
3	1401	2000
4	2001	3000
5	3001	9999

Examples of basic SELECT statements

- Select * from emp;
- Select empno from emp;
- Select empno,ename,job from emp;

Concatenation Operator (||)

- Concatenates columns or character strings to other columns
- Creates a resultant column that is a character expression
- `SELECT ename||job AS "Employees" FROM emp;`

Eliminating Duplicate Rows

- DISTINCT keyword is used for the elimination of duplicate rows in the output
- SELECT DISTINCT deptno FROM emp;

References and Additional Material

- Oracle 11g
- [Oracle Database Express Edition 11g Release 2](#)
- Apex Oracle
- apex.oracle.com